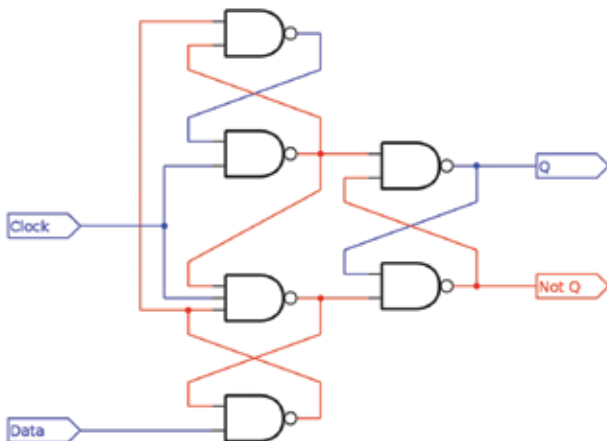


platform you use, so you need to pick one which suits your needs. The Raspberry Pi, for instance has a simulator called SmartSim, which can run natively on Raspbian and on other Operating Systems as well, where you can use the GPIO ports of the Raspberry Pi virtually, to test out your circuit logic.

The very popular Arduino has several tools online, which will help you get familiar with the workings of the various Arduino prototyping boards. AutoDesk, a widely popular company which designs software for makers and professionals alike, has an online tool called 123D Circuits [<http://dgit.in/123dcircuits>] for the Arduino and basic electronic circuits, which you can use to simulate your Arduino circuits. Although the components available are very basic and often limited to what is available in their own Arduino Starter Kit [<http://dgit.in/123dArduino>], it is still a good option if your circuit doesn't involve too many components. We had a bit of success designing a Obstacle Avoidance Robot using an ultrasonic sensor, though we had to replace the sensor with a push button switch in the simulator.

Apart from simulators, you can find circuit designers, which are basically tools that allow you to put circuits together and even design a PCB. Fritzing, an Open Source project, is one such software. Fritzing allows you to design Arduino and other electronics circuits, add external components

Positive Edge Triggered D Flipflop



A flip flop being tested on the SmartSim interface on a Raspberry Pi